CRM TEAM REPORTING · CHURN PREDICTION

Which customers should we prioritize for retention, and why?

What the churn model tells us about who is likely to stop buying, and where retention effort is worth the most



Based on sales data from Dec 2009 to Dec 2011

THE SHORT ANSWER

Prioritize customers by how much we stand to lose, not only by how likely they are to leave

15×

more lost spend found when the list is sorted by value at risk

£249,055 against £16,157, top 100 customers in each list

54%

of all value at risk sits with Steady Regulars and Fading Big Spenders

£325,952 of £602,796 in the holiday test group

2 models

one for the run-up to Christmas, one for the rest of the year

Using the wrong one misstates churn by about 10 percentage points

So what? The model improves on the RFM score by a modest, steady margin. Its biggest value to the team is showing where the money at risk sits.

WHAT WE MEAN BY CHURN

A customer has churned if they place no order in the 90 days after the cutoff date

Non-holiday season

Cutoff 10 Jun 2011

Everything the customer did before the cutoff

This is all the model is allowed to use

Next 90 days

10 Jun – 7 Sep 2011

68%

placed no order

Holiday season

Cutoff 9 Sep 2011

Everything the customer did before the cutoff

This is all the model is allowed to use

Next 90 days

9 Sep – 7 Dec 2011

57%

placed no order

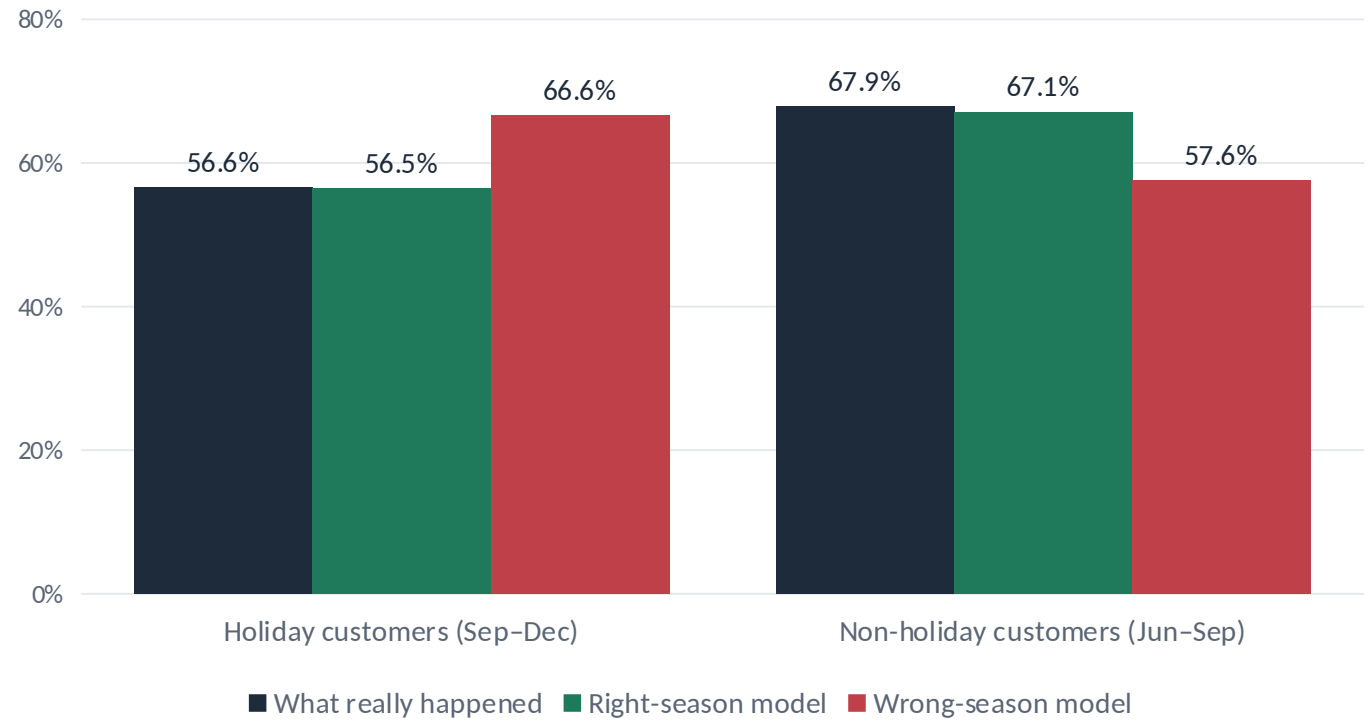
The model predicts churn separately for each season, from what we knew on the cutoff date. That is exactly how the team would use it on a real day.

So what? Churn is a clear, checkable event: no order in 90 days. Every figure in this deck uses that one definition.

WHY TWO MODELS

Customers behave differently before Christmas, so each season needs its own model

Share of customers expected to leave, against what really happened



About 10 points

The wrong-season model gets the overall churn level wrong. The summer model expects 67% to leave before Christmas; 57% did.

Same order, different level

Both models put customers in almost the same order from high to low risk. The difference is how much churn they expect in total.

So what? Use the holiday model for the September to December run-up and the non-holiday model outside it, or forecasts of lost customers will be off.

WHAT THE MODEL LOOKS AT

The model uses 16 details about each customer, which answer six plain questions

Recency and recent activity

6 details

How long since their last order? How much did they buy in the last 30, 90 and 180 days? Are they overdue against their usual rhythm?

Purchase frequency and spend

3 details

How many orders have they placed, how many different products have they bought, and how much have they spent in total?

Seasonal buying

1 detail

Did they order in the same 90 days one year earlier?

Customer history

1 detail

How long have they been a customer?

Order habits

3 details

How big is a typical order and basket, and how many days usually pass between orders?

Account details

2 details

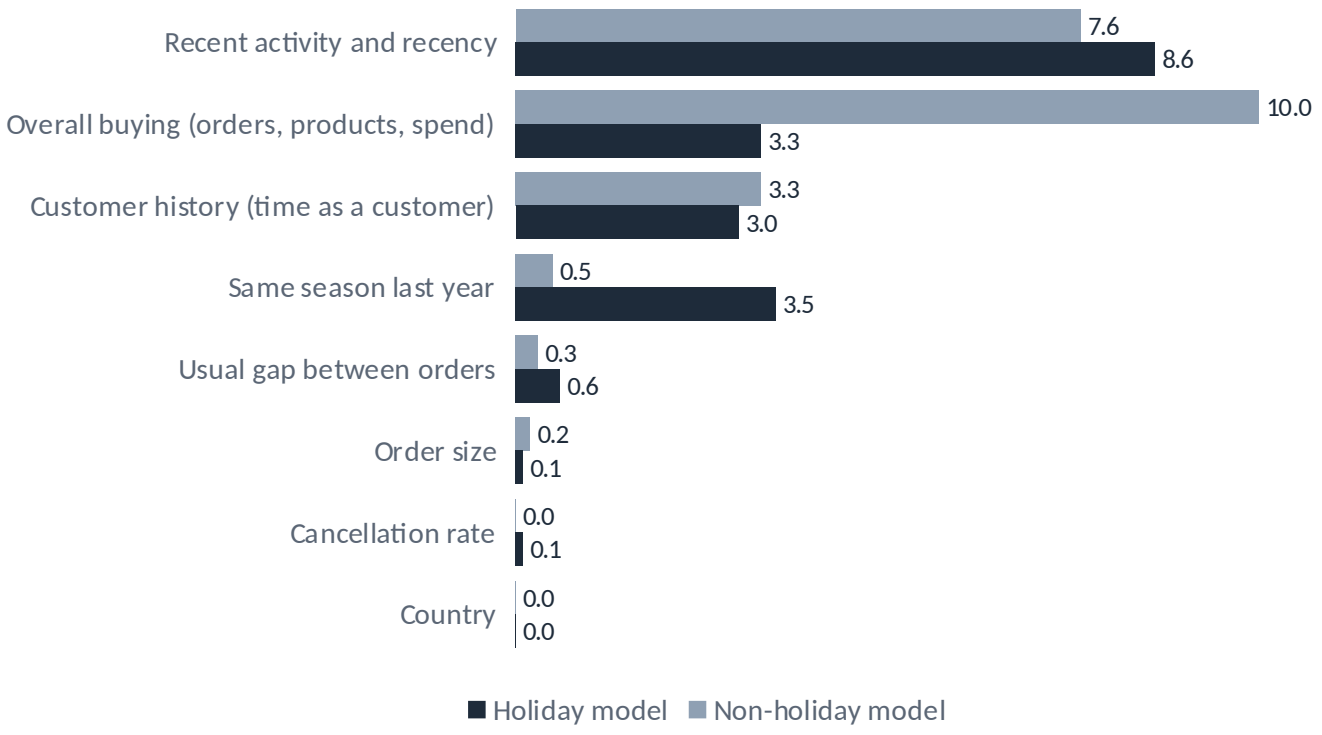
How often do they cancel, and are they based in the UK?

So what? Every one of these details is already in our sales records on the cutoff date. No new data has to be collected.

WHAT SIGNALS CHURN

Recent activity and overall buying tell us the most about who will leave

How much the model's ranking gets worse when each type of information is hidden (points out of 100)



Outside Christmas

Overall buying is the strongest signal: how many orders, how many products, how much spent.

Before Christmas

Recent activity is the strongest signal: when they last ordered and what they bought in recent months.

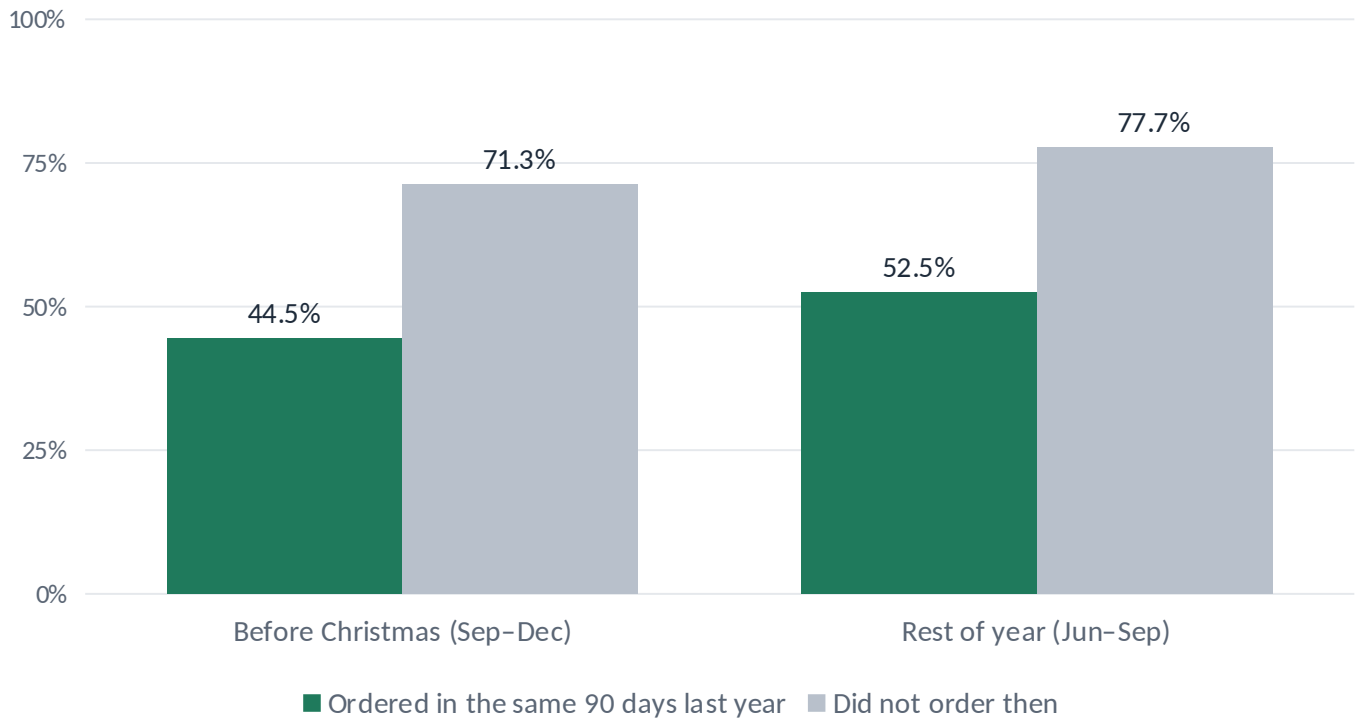
Order size, order gap, cancellations and country add almost nothing, so they are not a basis for action.

So what? A customer whose recent orders have dried up is the clearest warning sign, especially from September onwards.

THE SEASONAL SIGNAL

Last year's Christmas buying is one of the best guides to this Christmas

Share of customers who placed no order in the next 90 days



7×

more important in the holiday model than in the non-holiday model (3.5 points against 0.5)

Who to look for

Customers who ordered last September to December but have not ordered yet this autumn. Before Christmas, 55 in 100 of last year's buyers came back, against 29 in 100 of the rest.

So what? In early September, pull last year's Christmas buyers who have not ordered yet, and remind them before the rush.

HOW WELL IT WORKS

The model ranks customers better than the RFM score in both seasons, by a modest but steady margin

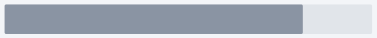
Pick one customer who left and one who stayed. How often does each method rank the one who left as riskier?

Non-holiday season

85 in 100
Churn model



81 in 100
RFM score



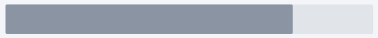
AUC 0.851 against 0.806

Holiday season

81 in 100
Churn model



78 in 100
RFM score



AUC 0.809 against 0.778

1,000 of 1,000

random re-checks in which the model stayed ahead of the RFM score

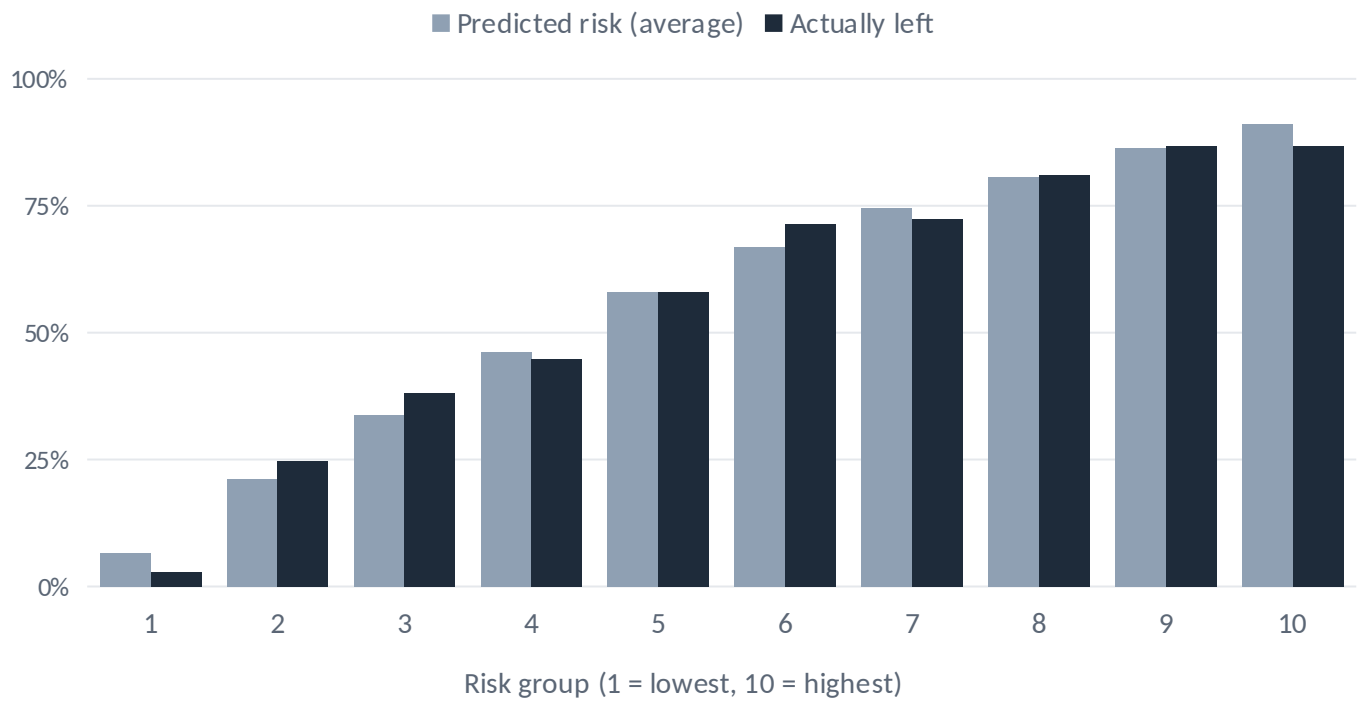
Tested on 2,040 customers the model had never seen. A coin flip would score 50 in 100.

So what? A modest, steady improvement, not a dramatic one. The model sharpens the RFM score; it does not replace what the team already uses.

READING THE RISK SCORE

A risk score is a likelihood, not a verdict

Holiday test customers, split into 10 equal groups from lowest to highest risk



The numbers can be read as they are

In every group, predicted and actual churn are within 5 points. When the model says 70%, about 70 in 100 such customers leave.

High risk is not certain

In the highest-risk group, 13 in 100 still came back. In the lowest, 3 in 100 still left.

A real example

Customer 12357 had one order of £11,230 and a 59% risk. They ordered again.

So what? Use the score to decide who to contact first, not to write customers off. Some high-risk customers still come back.

THE MAIN RECOMMENDATION

Rank customers by value at risk: the chance they leave multiplied by what they have spent

Chance of leaving	×	Past spend	=	Value at risk
from the churn model		total spend with us		what we stand to lose

Customer A Very likely to leave, but little to save	95%	×	£200	=	£190
Customer B More likely to stay, yet far more worth a call	50%	×	£20,000	=	£10,000

So what? A list sorted by risk alone fills up with customers who spent little and left long ago. Value at risk puts the money first.

THE EVIDENCE

The value at risk list finds 15 times more lost spend

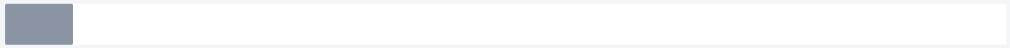
Top 100 by churn risk

87

of 100 left

£16,157

past spend among those who left



WHO IS ON THE LIST



Every one is a Gone Quiet customer

Typical customer: 1 order, £147 spent, last order 484 days before the cutoff.

Top 100 by value at risk

46

of 100 left

£249,055

past spend among those who left



WHO IS ON THE LIST



Fading Big Spenders 42 · Steady Regulars 32 · Top Customers 22 · Gone Quiet 4

Typical customer: 6 orders, £3,619 spent, last order 147 days before the cutoff.

So what? The churn risk list is right more often, but about customers worth little. The value at risk list is where retention effort pays.

BEHIND THE NUMBERS

Two real customers who both left: one had little to save, the other had spent almost £40,000

Customer 13438

Gone Quiet (Lapsed)

1	£46.50	455 days
order placed	spent in total	since last order

Churn risk **94%** Value at risk **£44**

8th on the churn risk list. The model was right: they did not order again. But there was very little to save.

Customer 15098

Steady Regulars (Loyal)

3	£39,620	90 days
orders placed	spent in total	since last order

Churn risk **55%** Value at risk **£21,632**

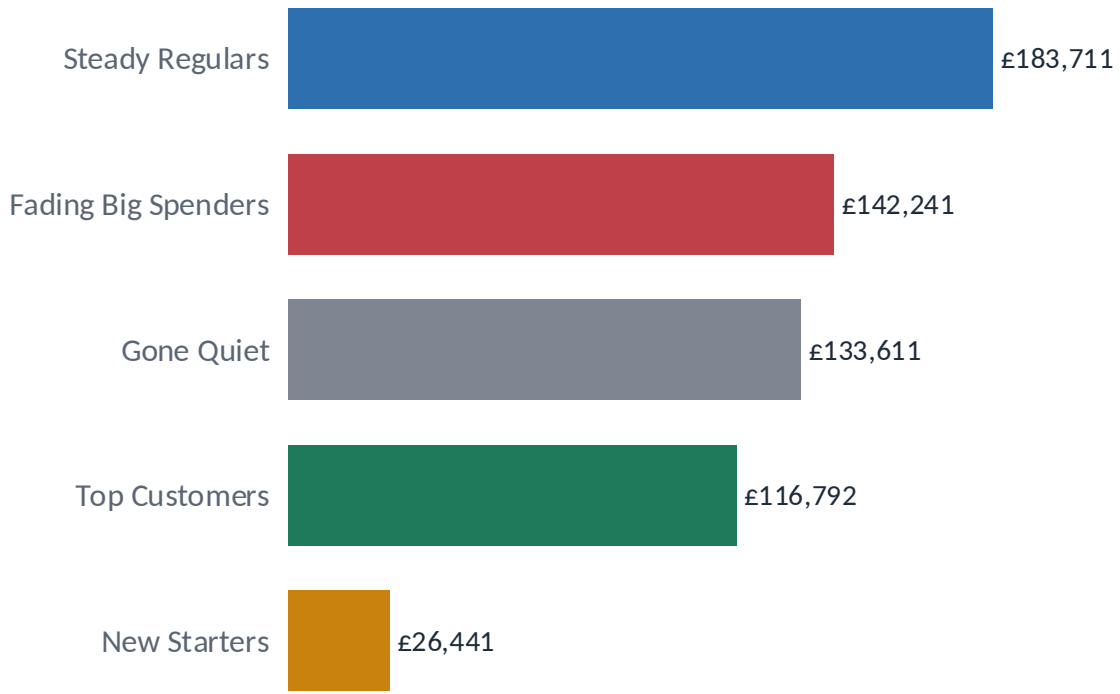
1st on the value at risk list, but outside the top 100 by risk. They did not order again in the run-up to Christmas.

So what? Customer 15098 alone had spent more than the whole top 100 of the risk-only list put together: £39,620 against £16,848.

WHERE THE VALUE AT RISK SITS

Steady Regulars and Fading Big Spenders hold the most value at risk

Total value at risk by customer group, holiday test group (1,051 customers)



Steady Regulars (Loyal)
38% average risk · £843 per customer
Moderate risk, but 218 customers who spend well: the largest total.

Fading Big Spenders (At Risk)
60% average risk · £1,580 per customer
Only 90 customers, but the most at risk per customer.

Gone Quiet (Lapsed)
81% average risk · £308 per customer
The highest risk, but spend is spread thin over 434 customers.

Top Customers (Champions)
15% average risk · £675 per customer
The lowest risk, but high spend still leaves £116,792 at risk.

So what? Put personal effort into Steady Regulars and Fading Big Spenders, watch Top Customers for early signs, and keep Gone Quiet contact low-cost.

WHAT TO DO · ORDERED BY VALUE AT RISK PER CUSTOMER

What the CRM team does with each group

Fading Big Spenders

AT RISK PER CUSTOMER

£1,580

Most value at risk per customer

- Personal call from the account team
- Offer based on their past favourite products
- Main push in September and October

Steady Regulars

AT RISK PER CUSTOMER

£843

Largest total value at risk

- Reminder when they are overdue against their usual rhythm
- Personal email for the highest value at risk

Top Customers

AT RISK PER CUSTOMER

£675

Low risk, high spend

- Flag any Top Customer whose risk rises
- Service and attention, not discounts

Gone Quiet

AT RISK PER CUSTOMER

£308

Highest risk, low spend

- Low-cost bulk email only
- One reminder to last year's Christmas buyers

New Starters

AT RISK PER CUSTOMER

£194

Low spend so far

- Automated welcome journey
- Aim for the third order

So what? The customer list does the sorting; these rules decide the type of contact. Personal effort goes where the money is.

PUTTING IT TO WORK

A simple routine each season, with the model as a prioritization tool



Run the right model

Early September for the Christmas run-up; the non-holiday model outside it



Sort by value at risk

Chance of leaving × past spend, highest first



Contact from the top

Type of contact follows each customer's group



Check after 90 days

Compare with what happened; refresh the model yearly

What the score is

A ranked list of where retention effort is most likely to pay off, based on how similar customers behaved in the past.

What the score is not

A promise that a customer will leave, or a reason to stop looking after customers with a low score.

So what? The routine is the same every season. Only the model and the cutoff date change.

WHAT TO KEEP IN MIND

The limits of this analysis, and what they mean for the team

Not yet tested on a later year

The model was checked on customers it had not seen, but from the same period. How it holds up in 2012 is still unknown.

Two years of data

Sales from December 2009 to December 2011 only, so each season could be tested once.

Results move a little

Each test group had about 1,000 customers, so the AUC results can move by about 0.02 (2 in 100) either way.

Favours long-standing customers

Value at risk uses total past spend, so older accounts can look bigger than their current buying.

Infrequent buyers can look churned

A customer who normally orders every four months can be counted as churned after 90 days, when they are only between orders.

None of these stop the team from using the list now. They are the first things to check as new sales data comes in.

So what? Treat the list as a strong starting point, check it every season, and improve it as new data arrives.

Prioritize by value at risk, use the right model for the season

What we learned

- Churn means no order in the 90 days after the cutoff. 57% left before Christmas, 68% in summer.
- Two seasonal models are needed; the wrong one misstates churn by about 10 points.
- Recent activity is the clearest warning sign, and last year's Christmas buyers are more likely to return.
- The model beats the RFM score modestly but consistently: 85 vs 81 and 81 vs 78 in 100.
- Sorting by value at risk finds 15 times more lost spend than sorting by risk alone.

Next steps

1. NOW — Add churn risk and value at risk to each customer record in the CRM
2. EARLY SEPT — Run the holiday model and contact the top of the value at risk list
3. EARLY SEPT — Remind last year's Christmas buyers who have not ordered yet
4. AFTER 90 DAYS — Compare the list with what happened, and record who responded
5. WITH 2012 DATA — Test on a later year, try recent 12-month spend, review the 90-day window

APPENDIX

Technical details for interested readers

A1 How the model was tested A2 The 16 customer details in full

Customers were split before any model was built, and the test customers were used only once

5,253 customers

Everyone who ordered before 9 Sep 2011

80% to learn from

4,202 holiday and 3,955 non-holiday customers

20% kept aside

1,051 holiday and 989 non-holiday customers, hidden from both models

Two model types were compared on the learning customers only. Gradient boosting scored 0.002 to 0.003 higher than logistic regression, less than the normal variation between checks, so the simpler logistic regression was chosen for both seasons.

On hidden customers	Non-holiday	Holiday
AUC, churn model	0.851	0.809
AUC, RFM total score	0.806	0.778
Accuracy (cut-off 50%)	80.1%	74.8%
Precision	82.7%	75.2%
Recall	89.4%	82.7%
F1 score	0.859	0.788

Accuracy looks high partly because most customers leave: a rule that says everyone leaves would already be 67.9% (non-holiday) and 56.6% (holiday) accurate. This is why ranking (AUC) is the main measure.

Precision: of customers flagged as leaving, the share who left.
Recall: of customers who left, the share flagged.

So what? Every result in this deck comes from customers the model never saw while it was being built.

Every detail is worked out from sales records before the cutoff date

Question	Name in the notebook	What it measures
Recency and recent activity	`recency`	Days since the last order
	`orders_last_30d / 90d / 180d`	Orders in the last 30, 90 and 180 days
	`spend_last_90d`	Spend in the last 90 days
	`recency_vs_gap`	Days away compared with their usual gap between orders
Purchase frequency and spend	`frequency`	Number of separate orders
	`product_breadth`	Number of different products bought
	`monetary`	Total spend after cancellations
Seasonal buying	`orders_same_season_last_year`	Orders in the same 90 days one year earlier
Customer history	`tenure`	Days since the first order
Order habits	`avg_order_value / avg_basket_size`	Typical value and size of an order
	`avg_gap_days`	Usual number of days between orders
Account details	`cancellation_rate / is_uk`	How often they cancel; based in the UK or not

Rows with two names count as two details (orders in the last 30, 90 and 180 days count as three), giving 16 in total. One-time buyers have no gap between orders; the model fills in a typical value and also records that it was missing.